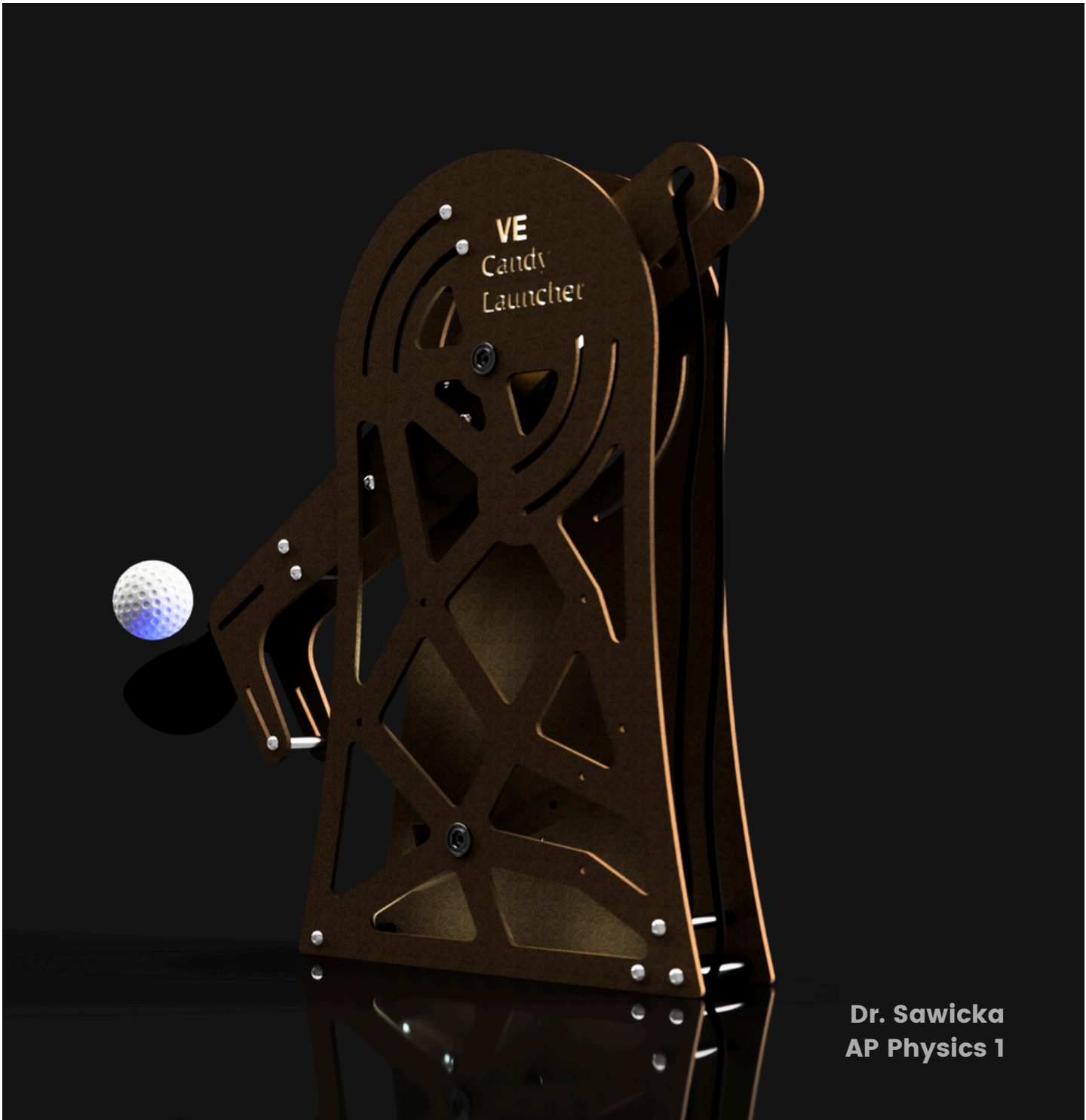
CANDY LAUNCHER LAB

By Eric Hou and Viviana Chen



Dr. Sawicka
AP Physics 1

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PREFACE

The objective of this lab is to build a “candy launcher” to launch an object in a targeted area that is 3 meters far and 1 meter above the launcher, by using what we’ve learned in the past semester.

By using real-life applications of projectile motion and work, we can calculate at what angle and initial velocity we must fire the object at using kinematic equations and formulas for spring potential energy and kinetic energy. Knowing these physical quantities between the launcher and the ball, we will be able to accurately fire the object into the targeted area.

We came up with multiple designs of a launcher, but our final design was the catapult.

Materials

- 3D prints
- laser-cut 1/8 in hardboard
- 8mm bearings
- standoffs
- screws
- surgical tubing



3. DESIGN PROCESS

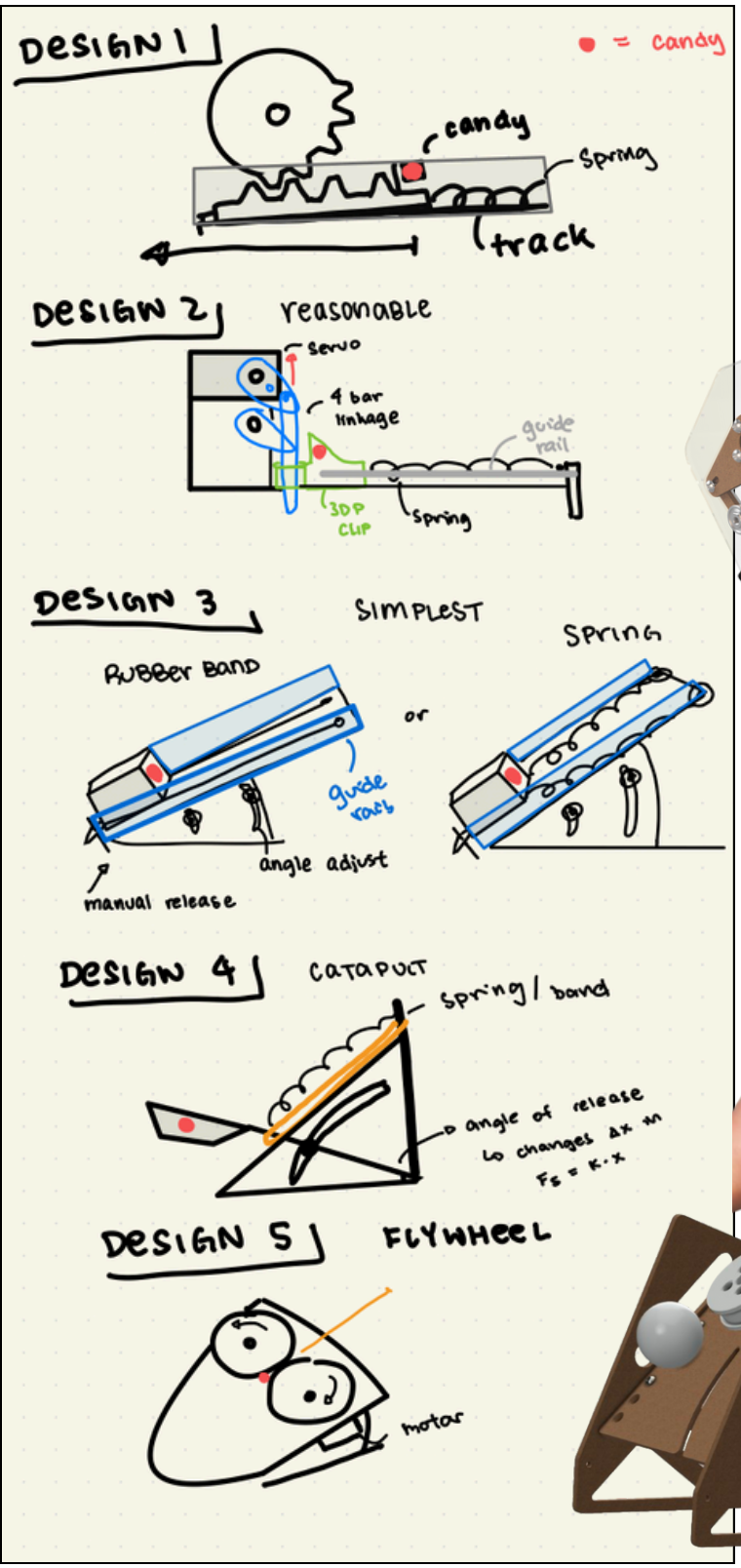
01 — Brainstorming & Prototyping

During the initial brainstorming phase, we scopes out ideas while analyzing advantages and potential drawbacks. We go beyond by developing proof of concepts and evaluating various designs. We arrive at the most optimal solution by incorporating intricate calculations and meticulously considering factors such as ease of use and cost. This gives us a good starting point where we can start doing calculations and making prototypes.

Mechanism	Pros	Cons
FLYWHEEL	• Consistent • Predictable • Adjustable	• Need to buy a motor • Needs electronics -> more time and **complexity**
RUBBER BAND / SPRING	• Easy to make adjustments	• Need to buy springs • Need a consistent release mechanism ◦ may require electronics => more time and complexity • More pull force
CATAPULT ***Winning Design**	• Simple & Reliable • Capable of Quick Adjustments • Easy mounting	• Less adjustability and precision maybe?

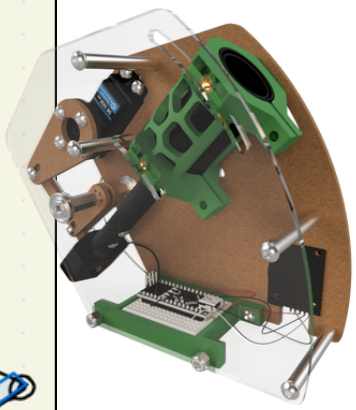
02 — Calculations and Research

Why we went with catapult? - Part 2



PROOF OF CONCEPTS

Helps us to identify design drawbacks and feasibility
Below are the different designs we tried out:



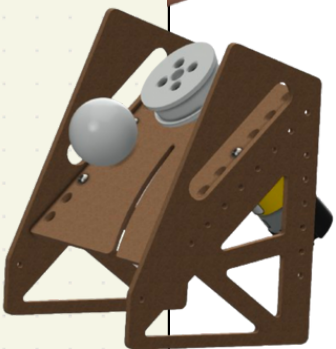
Compression Spring:
made the design for candy rather than golf ball



** Explanation on why we didn't proceed with design on the following page.

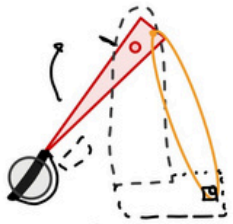


2nd Submission Catapult



Flywheel
**didn't finish design since we didn't want to include electronics since it would increase complexity

Final Design: Catapult.



Spring $KE = U_s$

$\frac{1}{2}mv^2 = \frac{1}{2}kx^2$

$v = \sqrt{\frac{kx^2}{m}} = x\sqrt{\frac{k}{m}}$

estimated spring val.

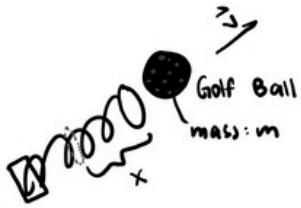
$k = 400\ N/m$

Spring constant $4.00\ N/m$ -----

Golf ball mass $0.046\ kg$ + other → $0.1\ kg$

$x = \sqrt{\frac{m}{k}} \cdot v$

$\sqrt{\frac{0.1}{400\ N}} \cdot 6.39\ m/s = 0.1\ m$



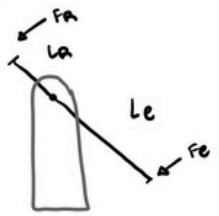
$F = -kx$

$F = (0.1)(400) = 40\ N!$

that's equivalent to 9 lb force to pull back over a distance of 10 cm!

Because it takes a significant amount of force to pull back, we can use mechanical advantage of a lever arm of a catapult

CATAPULT



ideal mechanical advantage

$IMA = Le/LR$

$IMA = \frac{FA}{Fe}$

$F_R = Ima \cdot Fe$

$Le = 300$

$LA = 80$

$IMA = \frac{300}{80} = 3.75$

$Ima = 3.75$ so $Fe = \frac{FA}{3.75}$

we can produce a force 3.75_x as powerful on the other side of the lever, which should help us to achieve the pull-back force nee ded.

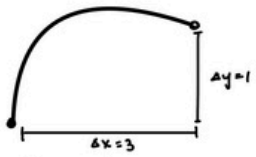
Angle & Velocity Derivations

TARGET

$\Delta y = 1$ m

$\Delta x = 3$ m

assume h=0



INITIAL ANGLE AND VELOCITY

$\Delta y = \frac{1}{2} a t^2 + v_y t$ $\quad v_y = v \sin\theta$

$\vec{a} = -9.81\ m/s^2$

$-4.905 t^2 + v_y t = 1$

$-4.905 t^2 + v_y t - 1 = 0$

$$\frac{-v\sin\theta \pm \sqrt{(v\sin\theta)^2 - 4(-4.905)(-1)}}{-9.81} = t$$

only counting the farther distance

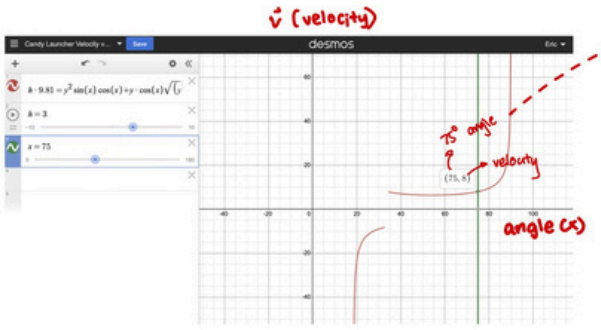
$$\frac{v\sin\theta + \sqrt{v^2\sin^2\theta - 2g}}{9.81} = t$$

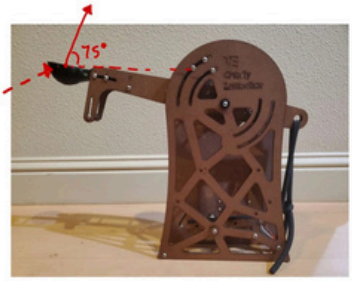
$\Delta x = v_x t$ $\quad v_x = v\cos\theta$

$$\Delta x = v\cos\theta \cdot \left(\frac{v\sin\theta + \sqrt{v^2\sin^2\theta - 19.62}}{9.81}\right)$$

$$\Delta x = \frac{v^2 \sin\theta\cos\theta + v\cos\theta\sqrt{v^2\sin^2\theta - 19.62}}{9.81}$$

let's say $\Delta x = 3$ m, $\Delta y = 1$ ← TARGET POSITION





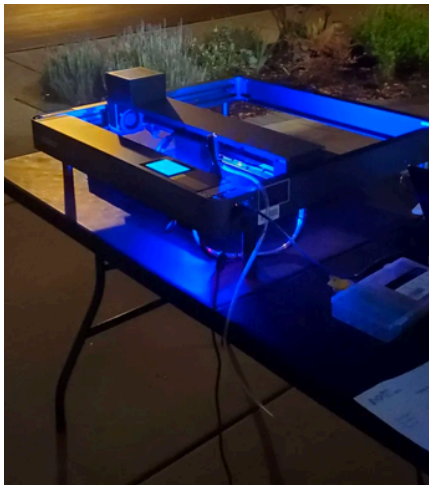
75° angle corresponds to 8 m/s

03 — Finalize and Test w/ Data

We built everything using a laser cutter as well as OTS parts, such as springs and surgical tubing



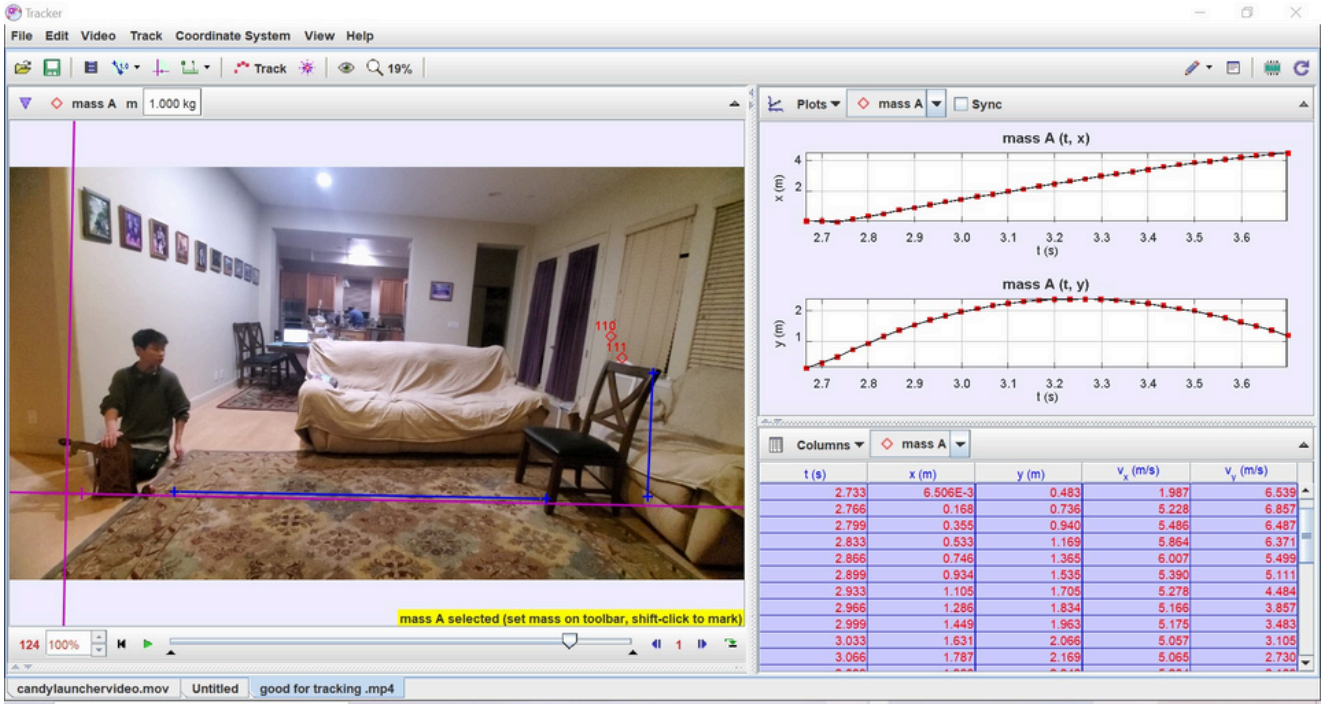
Laser-Cut Parts





Build From Parts

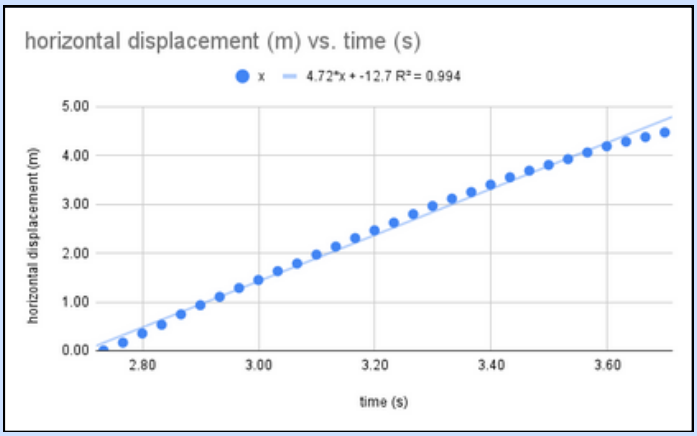
We used a physlet tracker to track the horizontal/vertical displacement and velocity and compared it to the actual values.

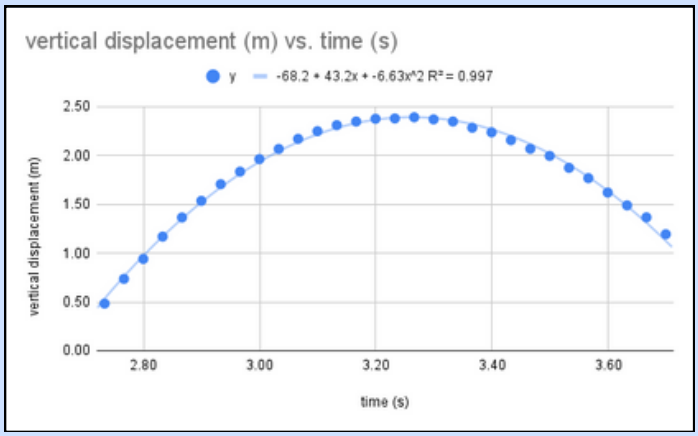


DATA

	mass_A				
t	x	y	vx	vy	initial velocity
2.73	0.01	0.48	5.32	6.54	8.43
2.77	0.17	0.74	5.23	6.86	
2.80	0.36	0.94	5.49	6.49	
2.83	0.53	1.17	5.86	6.37	
2.87	0.75	1.36	6.01	5.50	
2.90	0.93	1.54	5.39	5.11	
2.93	1.11	1.71	5.28	4.48	
2.97	1.29	1.83	5.17	3.86	
3.00	1.45	1.96	5.18	3.48	
3.03	1.63	2.07	5.06	3.10	
3.07	1.79	2.17	5.07	2.73	
3.10	1.97	2.25	5.20	2.11	
3.13	2.13	2.31	5.09	1.48	
3.17	2.31	2.35	4.98	0.98	
3.20	2.47	2.38	4.74	0.47	
3.23	2.62	2.38	4.99	0.23	
3.27	2.80	2.39	5.13	-0.14	
3.30	2.96	2.37	4.76	-0.65	
3.33	3.12	2.35	4.27	-1.28	
3.37	3.25	2.28	4.28	-1.66	
3.40	3.40	2.24	4.54	-1.90	
3.43	3.55	2.16	4.30	-2.53	
3.47	3.69	2.07	3.80	-2.42	
3.50	3.81	2.00	3.56	-2.92	
3.53	3.92	1.87	3.82	-3.42	
3.57	4.06	1.77	3.95	-3.79	
3.60	4.19	1.62	3.34	-4.18	
3.63	4.28	1.49	2.83	-3.81	
3.67	4.38	1.37	2.84	-4.44	
3.70	4.47	1.19			

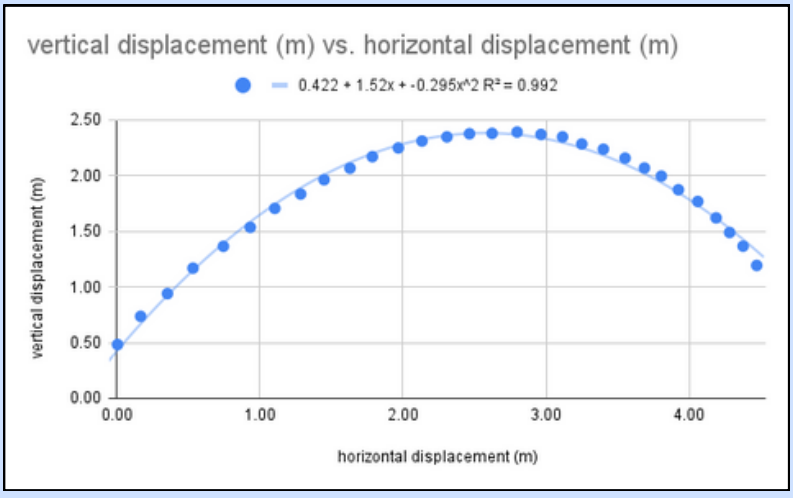
EVALUATION OF DATA





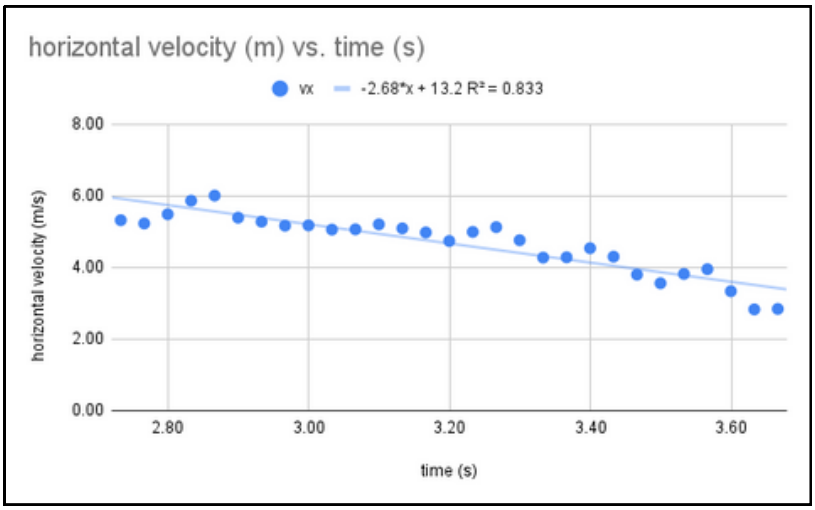
Equation: x=4.72t-12.7
the slope represents the change in horizontal distance over change in time

Equation: y=-6.63t^2 +43.2x-68.2

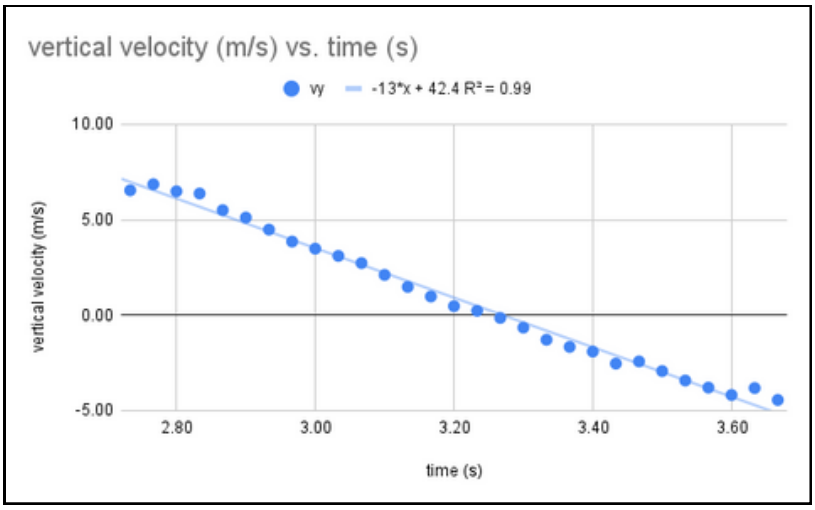


Equation: y=-0.295x^2+1.52x+0.422





Equation: vx = -2.68x+13.2
the slope represents the acceleration of the object in the x direction.



Equation: vy = -13x+42.4
the slope represents the acceleration of the object in the y direction, which is gravity.

PERCENT DIFFERENCE

This is the percent difference between our calculated velocity and actual velocity of the object we got from the tracker data. This difference makes sense because we overshot the target by a small amount.

Solution:

Calculate percentage difference between $V_1 = 8$ and $V_2 = 8.43$

$$\frac{|V_1 - V_2|}{\left[\frac{(V_1 + V_2)}{2}\right]} \times 100 = ?$$

$$= \frac{|8 - 8.43|}{\left[\frac{(8 + 8.43)}{2}\right]} \times 100$$

$$= \frac{|-0.43|}{\left[\frac{16.43}{2}\right]} \times 100$$

$$= \frac{0.43}{8.215} \times 100$$

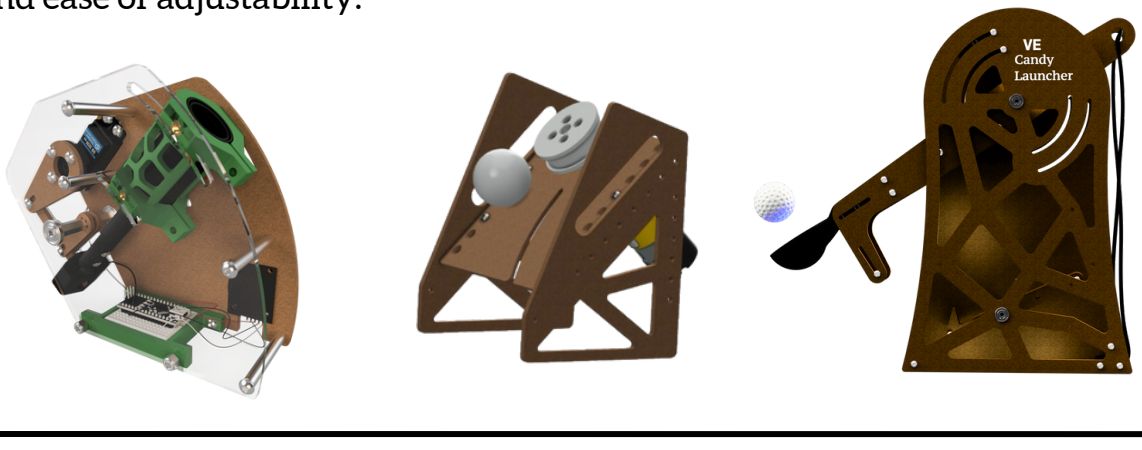
$$= 0.0523433 \times 100$$

$$= 5.23433\% \text{ difference}$$

5.23%

CONCLUSION

Our design process was a journey through three distinct stages: Brainstorming & Prototyping, Research & Calculations, and finally, Building & Testing. We explored various design options, starting with a spring launcher, moving on to a flywheel design, and ultimately settling on a catapult design. This iterative process taught us the importance of applying calculations and derivations to determine the factors influencing our launcher, eliminate less effective design options, and calculate our initial velocity. The final design we chose incorporated surgical tubing and a lever arm catapult. This choice was driven by its simplicity in construction, reliability, and ease of adjustability.



Data collection was carried out using a physics tracker and video recording. The horizontal displacement showed a fairly linear trend with a slope of 4.72 m/s, corresponding to our horizontal velocity. There was also a slight downward trend in the horizontal velocity graph, which could be attributed to non-conservative forces such as air resistance and potential inaccuracies due to the camera angle.

For our vertical displacement, we had a downward parabola, which had an equation of -6.63x^2 + 43.2x - 68.2. Our vertical velocity was a negatively sloped line, which had an equation of -13x + 42.4. This is logical as the vertical velocity decreased from the acceleration due to gravity.

We calculated our initial velocity by evaluating $\sqrt{Vxi^2 + Vyi^2}$, which returned a value of 8.43 m/s. The percent difference between our predicted value of 8 m/s was 5.23%, which is reasonable as we slightly overshot our target of 3 m.

POSTLAB NOTES

General Equations:

Kinematics

$$\triangle x = \frac{1}{2}at^2 + vt$$

Potential Energy

$$PE = \frac{1}{2}kx^2$$

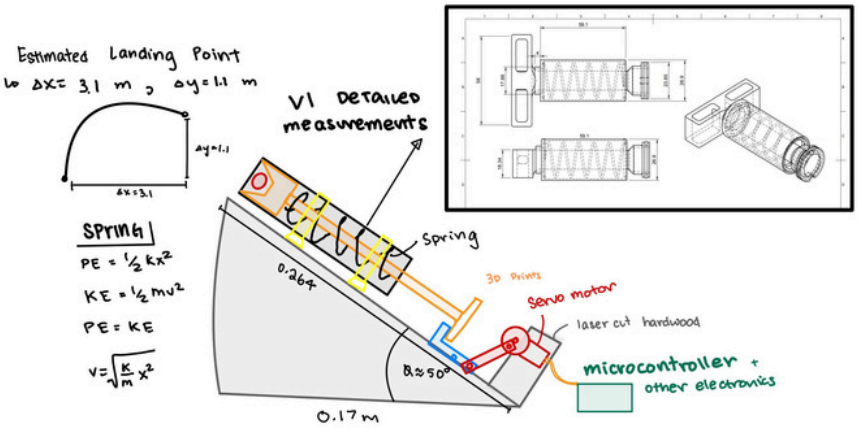
Kinetic Energy

$$KE = \frac{1}{2}mv^2$$

$$PE = KE$$

$$v = \sqrt{\frac{k}{m}x^2}$$

Other Designs:



materials
- spring
- servo
- microcontroller
  - button / trigger
- PLA / PETG (3D print)
- hardboard (laser cut)
- metal linkages?
- heat set inserts

The rest of the measurements (involving the release mechanism) do not affect the trajectory of the candy so can be changed throughout testing.

